

# ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

## TRAINING TR-102 REPORT DAY 7

1 August 2025

### Overview:

We had a packed and interesting day where we explored multiple topics. First, we learned about Vim, a powerful text editor, including how to install it and basic commands to use it. Then we were introduced to Kaggle, a popular platform for data science and machine learning projects. After that, we revised the basics of Python programming — including syntax, how to write variables, add comments, use type casting, and understand different data types.

### Learning Objectives:

- Get familiar with Vim editor and its workflow.
- Learn how to install and use Vim for editing code.
- Understand what Kaggle is and how to use it.
- Revise Python basics: syntax, variables, comments, casting, and data types.

### Introduction on Vim and Workflow:

Vim is a powerful and lightweight text editor that runs in the command line. It is widely used by developers and system administrators because of its speed, flexibility, and keyboard-based workflow. Unlike modern editors with graphical interfaces, Vim may seem confusing at first, but once you get used to it, it can greatly improve productivity.

Vim has different modes, the most important ones being Normal Mode (for navigation and commands) and Insert Mode (for writing text). To start typing, you press `i` to enter Insert Mode, and once you're done, press `Esc` to go back to Normal Mode.

### Basic commands:

- `:w` to save the file,

- :q to quit,
- :wq to save and quit,
- and :q! to quit without saving.

## Introduction on Kaggle:

**Kaggle** is a popular online platform for data science, machine learning, and artificial intelligence projects. It is a place where learners, professionals, and researchers come together to work on datasets, solve real-world problems, and take part in competitions. Kaggle provides free access to thousands of public datasets and Jupyter Notebook environments, so you can start coding right in the browser without setting up anything on your computer.

Kaggle is best in its community. We can learn from others by exploring their code, sharing your own work, and participating in discussions and forums. Kaggle also has competitions where companies and researchers post real challenges, and participants from around the world compete to solve them.

## Python Basics:

Python is a simple and beginner-friendly programming language known for its clean and readable syntax. It uses indentation (spaces or tabs) instead of curly braces to define code blocks, which makes the code easy to read and understand.

### Writing First Python Program

```
print("Hello World, Welcome to Python Basics")
```

### Output

```
Hello World, Welcome to Python Basics
```

## Comments in Python:

Comments in Python are the lines in the code that are ignored by the interpreter during the execution of the program. Also, Comments enhance the readability of the code and help the programmers to understand the code very carefully.

```
# This is a single-line comment
```

```
""" This is a multi-line comment or docstring. """
```

## Type Casting

Type Casting is the method to convert the Python variable datatype into a certain data type in order to perform the required operation by users. There are two types of Type Casting in Python:

- Python Implicit Type Conversion
- Python Explicit Type Conversion

### Implicit Type Conversion in Python

In this, method, Python converts the datatype into another datatype automatically. Users don't have to involve in this process.

### Explicit Type Conversion in Python

In this method, Python needs user involvement to convert the variable data type into the required data type.

### Examples of Type Casting in Python

Mainly type casting can be done with these data type functions:

- **Int():** Python `Int()` function take float or string as an argument and returns int type object.
- **float():** Python `float()` function take int or string as an argument and return float type object.
- **str():** Python `str()` function takes float or int as an argument and returns string type object.

### Example:

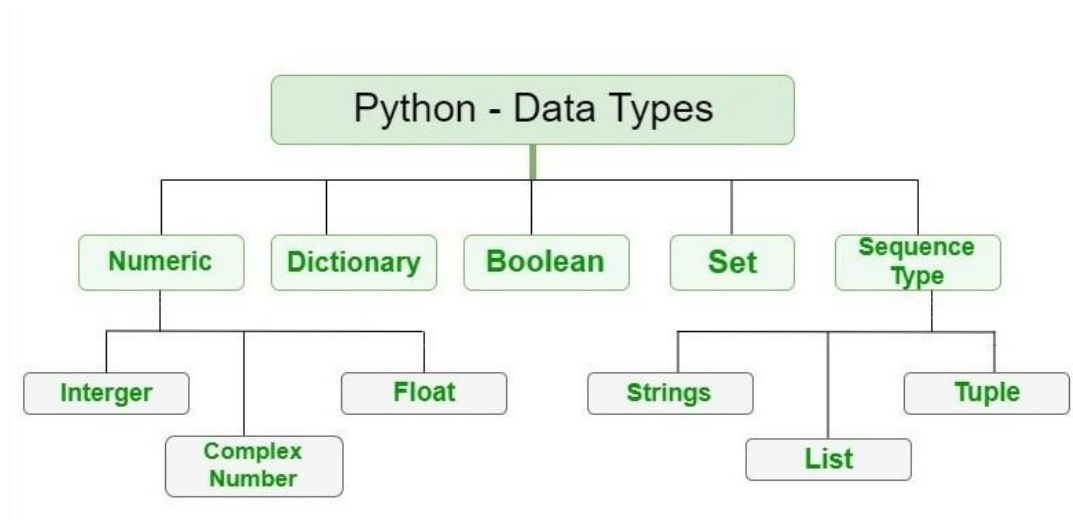
```
name = input("Enter your name: ")
print(type(name))

age = int(input("Enter your age: "))
print(type(age))
```

### Output:

```
Enter your name: Jeeya Thapar
<class 'str'>
Enter your age: 20
<class 'int'>
```

## Data Types in Python:



**Example:** This code assigns variable 'x' different values of various data types in Python.

```
x = "Hello World" # string
x = 50 # integer
x = 60.5 # float
x = 3j # complex
x = ["Jeeya", "for", "Thapar"] # list
```

```

x = ("Jeeya", "for", "Thapar") # tuple
x = {"Jeeya": "Thapar", "age": 20} # dict
x = {"Jeeya", "for", "Thapar"} # set
x = True # bool
x = b"Jeeya" # binary

```

## Python Input/Output

Python provides simple functions for input and output operations.

### Input

The **input()** function allows user input, example:

```

val = input("Enter your value: ")
print("You entered:", val)

```

### Output:

```

Enter your value: 11
You entered: 11

```

The **input()** function in Python always returns data as a string, regardless of what the user enters. If we want to store the input as another data type (like int, float, etc.).

## Conclusion:

Learning Vim was a bit tricky at first but fun once the basic commands were understood. Kaggle looks like a very useful platform for exploring real data projects. Revising Python basics was helpful to strengthen the foundation. Overall, it was a good mix of tools, platforms, and programming concepts.